EX. NO : 1
DATE : 06/08/26

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

AIM:

To study various network commands used in Linux and Windows:

NETWORK COMMANDS IN WINDOWS:

1) arp - Address Resolution Protocol



2) hostname , ipconfig



3) net stat



4) nb stat



nslookup :



pathping :



ping :



route :



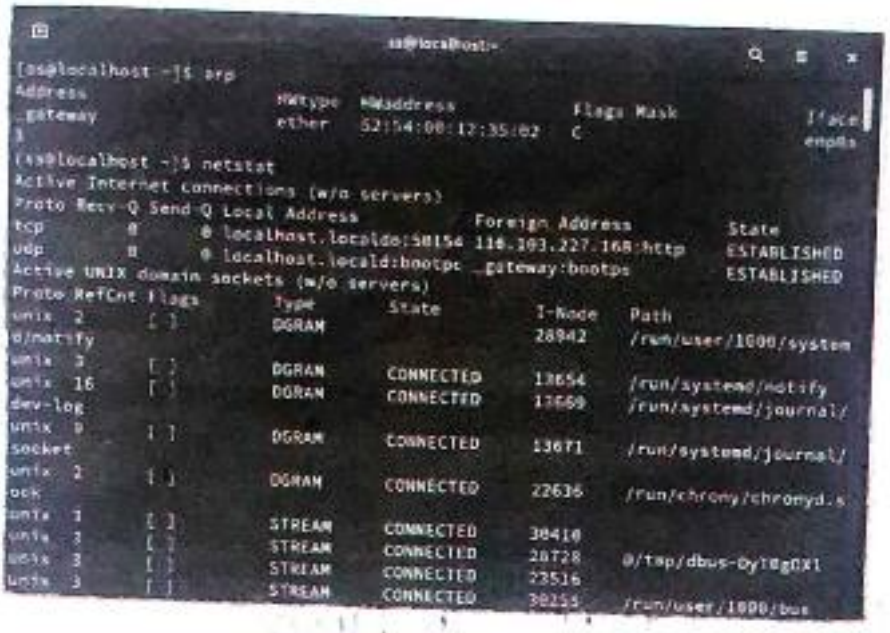
NETWORK COMMANDS IN LINUX

hostname

ip addr



arp , netstat



ping google.com



ss



tracepath google.com



nslookup google.com



Basic Networking commands

1) arp -a : ARP is a short form of address resolution protocol, It will show the IP address of your computer along with IP & MAC address of your router

2) hostname : This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

3) Ipconfig/all : This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP & type of Ethernet adapter in your system.

4) nbstat -a: This command helps solve problems with Net BIOS name resolution. (NBT stands for Net BIOS over TCP/IP.

5) netstat : (network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming & outgoing as well as viewing routing tables, & other Interface Statistics etc.

6) nslookup : (name server lookup) is a tool used to perform DNS lookups on Linux. It is used to display DNS details such as IP address of a particular computer.

nslookup can operate. In two modes : Interactive and non Interactive.

eg: ns lookup www.google.com

7) pathping : Pathping is unique to Windows and is basically a combination of the Ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each routers, gathers statistics on the rate of data loss along each hop.

8) ping: (Packet Internet Grouper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to other devices.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)

9) Route : route command is used to show/ manipulate the IP routing table. It is primarily used to setup static routers to specific host or network via an interface.

LINUX NETWORKING COMMANDS:

1) ip : The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems & assigning IP's to troubleshooting systems. The ip command can show address information, manipulate routing, display network various device, interface & tunnels.

2) ipconfig : The ipconfig command was/is a staple in many sys admin's tool belt for configuring & trouble shooting networks. It has since been replaced by the ip command.

3) mtr : MTR is a program with a command line interface that serves as a network diagnostic and trouble shooting tool combines the functionality of ping & trace route commands.

4) tcp dump : The tcp dump command is designed for capturing and displaying packets.

5) Ping : Ping is a tool that verifies IP level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request message. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability & name resolution.

RESULT:

Thus, the Linux and Windows networking commands has been executed successfully.